

1. Use the **limit of Riemann sum definition** the definite integral to evaluate $\int_0^2 (2 - 3x^2) dx$.

Note: You may use right-endpoint Riemann sum. No credit will be given if you use other method.

Formulas you may need: $\sum_{i=1}^n c = cn$, $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$,

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

2. Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of $F(x) = \int_0^{\ln x} \sqrt{t^3 + 1} dt$.